Mario Level Generation Guided by Pairwise Fun Preferences

CMPUT 652 Project Proposal

1 Research Objective

This project uses human fun preferences to guide procedural generation of 2D Super Mario Bros.-style levels. A convolutional variational autoencoder (CNN-VAE) reconstructs levels with their recorded paths. A fully connected layer (FC-Tail) learns which playthrough a participant preferred and guides a latent-space search for new levels.

2 Dataset

The project uses the cleaned Mario-Fun-Study release from Wang et al. [1]. Preprocessing retains 90 strict preferences from Experiment I and excludes 17 equally fun and 17 neither-fun responses. Each of the 91 ordered triples from Experiment II yields three comparisons, for 273 derived pairs. The resulting 363 comparisons cover 450 distinct participant–level playthroughs, 290 maps, and 84 participants. The labels express a relative preference within each comparison.

The supplied 80/20 split (seed 42) contains 290 training and 73 test comparisons. It keeps pairs from the same annotation, repeated rankings, and reused playthroughs together. The same recorded playthrough cannot appear in both splits, although some participants and maps appear in both.

3 Proposed Methodology

3.1 Level and Path Representation

Each playthrough uses a $22 \times 16 \times 192$ multi-hot tensor covering the uncropped level. The first 21 channels encode one tile type per cell. Recorded positions are mapped to grid cells without interpolation, skipping positions outside the grid. The path is encoded as a 22nd tile channel. Each cell has a geometry bit and gains a second active bit if the player visited it.

3.2 Shared CNN-VAE and Pairwise Prediction

The CNN-VAE processes one playthrough at a time. For each ordered pair (x_a, x_b), the same encoder E_ϕ produces two diagonal Gaussian posteriors. It draws reparameterized samples $z_i = \mu_i + \sigma_i \odot \epsilon_i$, with $\epsilon_i \sim \mathcal{N}(0, I)$. The shared decoder D_θ reconstructs each input from its own sample. The FC-Tail receives the two samples concatenated in input order. A sigmoid converts its output to $p(z_a, z_b)$, the predicted probability that the first playthrough is preferred to the second.

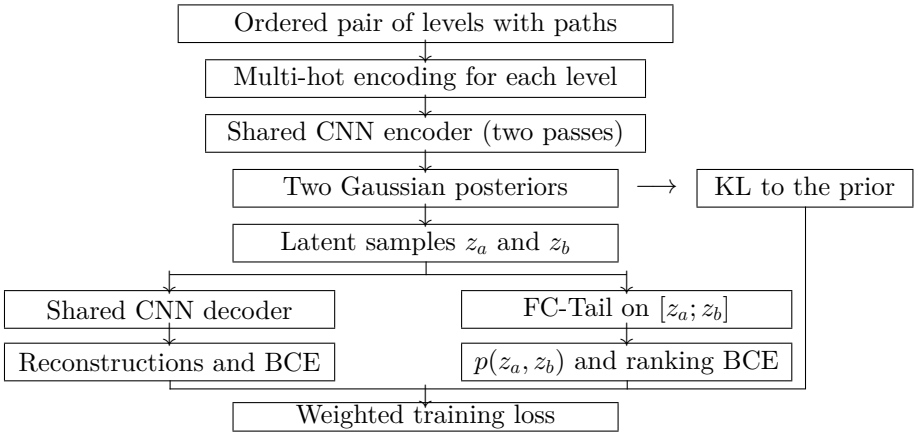

Figure 1: Training flow for the shared CNN-VAE and the pairwise FC-Tail.

3.3 Joint Training

If A was preferred to B, training presents (A, B) with target $y = 1$ and (B, A) with $y = 0$. Reversing the pairs after splitting gives 580 ordered training examples from the 290 underlying comparisons. The network minimizes

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{rank}}\,\text{BCE}(p(z_a, z_b), y). \tag{1}$$

One binary cross-entropy reconstruction loss averages over all 22 channels and all cells in both inputs. The decoder emits one logit per bit, so the tile and path bits can be active together. The KL term averages both posteriors' divergences from the standard normal prior. Ranking gradients pass through both reparameterized samples to the shared encoder. Adam updates all model parameters. The three loss weights and Adam learning rate will be selected on a validation subset that keeps the training comparison groups intact.

3.4 Level Generation by Latent Optimization

After training, freeze the model and draw two latent vectors from the VAE prior. Compare them in a fixed input order. Use the first vector as the initial reference if the predicted probability exceeds 0.5, and the second otherwise. Then repeat the following procedure for n rounds.

1. Draw a new candidate latent vector c from the prior. Pass it to the FC-Tail as the first input and the current reference r as the second.
2. Set the target to 1 and take a fixed number of gradient descent steps on $\text{BCE}(p(c, r), 1)$. Backpropagate through the frozen FC-Tail, updating only c while holding r fixed.
3. If $p(c, r) > 0.5$ after optimization, use c as the reference for the next round. Otherwise keep r.

Decode the final reference after round n. Select the highest-scoring geometry tile per cell from the first 21 channels. The path channel can be displayed as an overlay.

4 Preliminary Evaluation Plan

Compare guided generation with unguided prior sampling and with decoding the initial reference.

Fun ranking. Report accuracy and macro-F1 over both orders of the 73 held-out comparisons, reusing each pair of latent samples.

Playability. Report the fraction of generated tile maps an A* gameplay agent completes under a fixed budget.

Novelty. Report Average Minimum Normalized Hamming Distance. For each generated map, find the smallest fraction of differing tile cells over all training maps. Average these minima across generated maps.

Diversity. Use Expressive Range Analysis to compare Linearity vs. Leniency, Enemy Count vs. Enemy Sparsity, and Enemy Count vs. Enemy Vertical Spread across generation methods.

5 Expected Outcome

The project will assess whether a shared CNN-VAE can reconstruct Mario playthroughs and predict held-out fun preferences, and whether guiding generation with those predictions preserves playability, novelty, and variety.

References

[1] Z. Wang, Y. Li, H. Du, J. Liu, and G. N. Yannakakis. *Fun as Moderate Divergence: Evaluating Experience-Driven PCG via RL.* IEEE Transactions on Games, 17(2), 360–373, 2025. doi:10.1109/TG.2024.3456101.